

Two Closed Subspace-Cones Refute Both Lipschitz Klee–Andô Extensions

Candidate solution packet for arXiv:1606.00249

9 August 2026

Status. Candidate counterexample, likely valid: a full negative answer to Problem 1.5 of arXiv:1606.00249, subject to human review.

1 Source question

Messerschmidt’s paper [1] proves continuous positively homogeneous selection theorems for arbitrary collections of closed cones. On page 3 it asks:

Problem 1.5. Do Corollaries 5.5 and 5.6 remain true when, in their statements, the word “continuous” is replaced by the word “Lipschitz”?

Here Corollary 5.5 is the coadditivity theorem and Corollary 5.6 is the conormality theorem. The source defines a cone as a nonempty set closed under addition and multiplication by nonnegative scalars; properness is a separate condition. Thus every linear subspace is a cone.

both our Strong Klee-Andô Theorems for both coadditivity and conormality:

Theorem 4.5 (Open Mapping Theorem for cone-valued correspondences). *Let C a complete metric cone (as defined in Section 4), Y a Banach space and $D \subseteq Y$ a closed cone. Let $T : C \rightarrow Y$ be a continuous additive positively homogeneous map. If the correspondence $\Psi : C \rightrightarrows Y$ defined by $\Psi(c) := Tc + D$ ($c \in C$) is surjective (in the sense that, for every $y \in Y$, there exists some $c \in C$ such that $y \in \Psi(c)$), then the image under Ψ of the open unit ball about zero in C is open in Y .*

We point out the following problem toward further strengthening of the Klee-Andô Theorems proven in this paper. This problem is motivated by the observation that the lattice operations in every Banach lattice are Lipschitz [13, Proposition II.5.2], and the further question as to whether a form of this carries over to ordered Banach spaces, or more generally, to Banach spaces endowed with arbitrary collections of cones. To the author’s knowledge, this problem remains open:

Problem 1.5. *Do Corollaries 5.5 and 5.6 remain true when, in their statements, the word “continuous” is replaced by the word “Lipschitz”?*

A straightforward application of [12, Corollary 4.6] shows that Corollaries 5.5 and 5.6 remain true when, in their statements, the word “continuous” is replaced

Figure 1: Problem 1.5 in [1, p. 3].

2 Definitions and the quotient obstruction

Let $q : E \rightarrow Y$ be a bounded linear surjection of real Banach spaces. A right inverse is a map $s : Y \rightarrow E$ satisfying $q \circ s = \text{id}_Y$; it need not be linear. We use the classical fact that

$$q : \ell_\infty \longrightarrow \ell_\infty / c_0$$

has no uniformly continuous right inverse, and hence no Lipschitz right inverse. This is Kalton's example, recorded as Proposition 2.5 (and again in Corollary 5.9) of [2].

For two cones $C_1, C_2 \subseteq X$, the conormality selection requested by the Lipschitz version of Corollary 5.6 consists of Lipschitz maps $\delta_i : X \rightarrow C_i$ with

$$x = \delta_1(x) + \delta_2(x).$$

The coadditivity selection requested by the Lipschitz version of Corollary 5.5 is a Lipschitz map $v : X^2 \rightarrow X$ such that

$$v(\xi_1, \xi_2) \in (\xi_1 + C_1) \cap (\xi_2 + C_2).$$

For a two-point index set the source's spaces $c(\Omega, X)$ and $\ell_\infty(\Omega, X)$ both equal X^2 , with the maximum norm.

3 Proof intuition

The graph of a quotient map turns lifting into cone geometry. Decomposing $(0, y)$ into a horizontal vector and a graph vector necessarily chooses an $e \in E$ with $q(e) = y$. Similarly, an intersection point between a horizontal translate and a graph translate chooses such an e , up to sign. Therefore a Lipschitz selector in either formulation would give a Lipschitz right inverse of q . Choosing a quotient known not to have such a right inverse produces one pair of closed cones that defeats both proposed extensions.

4 Counterexample theorem

Theorem 1. *There exist a Banach space X and two closed cones $C_1, C_2 \subseteq X$ for which all of the following hold.*

1. *Every $x \in X$ can be written $x = c_1 + c_2$ with $c_i \in C_i$.*
2. *For every $(\xi_1, \xi_2) \in X^2$, the intersection $(\xi_1 + C_1) \cap (\xi_2 + C_2)$ is nonempty.*
3. *There are no Lipschitz maps $\delta_i : X \rightarrow C_i$ satisfying $x = \delta_1(x) + \delta_2(x)$ for every $x \in X$.*
4. *There is no Lipschitz map $v : X^2 \rightarrow X$ satisfying $v(\xi_1, \xi_2) \in (\xi_1 + C_1) \cap (\xi_2 + C_2)$ for every $(\xi_1, \xi_2) \in X^2$.*

Consequently, both clauses of Problem 1.5 in [1] have negative answers. The failure persists even if positive homogeneity and the quantitative norm bounds are omitted from the desired conclusions.

Proof. Let $q : E \rightarrow Y$ be any bounded linear surjection having no Lipschitz right inverse. Equip

$$X := E \oplus_\infty Y$$

with its maximum norm, and define

$$M := E \oplus \{0\}, \quad N := \text{graph}(q) = \{(e, qe) : e \in E\}.$$

Both M and N are closed linear subspaces of X , hence closed cones in the sense of [1]. Set $C_1 = M$ and $C_2 = N$.

First, $M + N = X$. Indeed, for $(a, y) \in X$, choose $e \in E$ with $qe = y$. Then

$$(a, y) = (a - e, 0) + (e, qe) \in M + N.$$

This proves assertion 1. It also proves assertion 2: given $\xi_1, \xi_2 \in X$, write $\xi_2 - \xi_1 = m + n$ with $m \in M$ and $n \in N$. Then

$$\xi_1 + m = \xi_2 - n \in (\xi_1 + M) \cap (\xi_2 + N).$$

Suppose assertion 3 fails. Thus Lipschitz maps $\delta_M : X \rightarrow M$ and $\delta_N : X \rightarrow N$ satisfy

$$z = \delta_M(z) + \delta_N(z) \quad (z \in X).$$

For $y \in Y$, write

$$\delta_N(0, y) = (s(y), y).$$

The second coordinate is y because the second coordinate of $\delta_M(0, y)$ is zero. Since $\delta_N(0, y) \in N$, we have $q(s(y)) = y$. Moreover s is Lipschitz, being the composition of the bounded embedding $y \mapsto (0, y)$, the Lipschitz map δ_N , and the bounded first-coordinate projection. Thus s is a Lipschitz right inverse of q , a contradiction.

Suppose assertion 4 fails, and let v be the asserted Lipschitz selector. For $y \in Y$, the first membership condition gives

$$v(0, (0, y)) \in M,$$

so write $v(0, (0, y)) = (a(y), 0)$. The second membership condition gives

$$(a(y), -y) = v(0, (0, y)) - (0, y) \in N.$$

Hence $q(a(y)) = -y$. The map $s(y) := -a(y)$ is Lipschitz and satisfies $q(s(y)) = y$, again contradicting the choice of q .

Finally take $E = \ell_\infty$, $Y = \ell_\infty/c_0$, and let q be the quotient map. By [2, Proposition 2.5], this q has no uniformly continuous right inverse, so in particular no Lipschitz right inverse. The preceding construction proves all four assertions. Neither contradiction used positive homogeneity or a norm estimate. $\square$

5 Verification notes and scope

The proof is structural and uses no computation. The points most useful to check are:

- N is closed because it is the graph of a bounded linear map;
- subspaces qualify as cones under the source definition;
- restricting either hypothetical selector to the displayed copy of Y really produces a right inverse of q ;
- a two-element index set simultaneously covers both allowed choices of Z in Corollary 5.5.

This theorem settles Problem 1.5 as stated, because that problem concerns the source's arbitrary-collection results. It does not settle the separate, stronger Conjecture 1.6, which requests one closed generating cone and Lipschitz positive and negative parts for that single cone.

6 Bounded novelty check

On 9 August 2026 we searched the run indexes, the source text, exact phrases from Problem 1.5, and close variants involving Klee–Andô selections, Lipschitz decomposition, and quotient right inverses. We also checked the 2019 survey on the single-cone Lipschitz decomposition problem and arXiv:1909.10417. We found the known quotient obstruction used above, but no source stating the two-subspace graph reduction or claiming a negative answer to both Corollaries 5.5 and 5.6. This is a bounded novelty check, not an exhaustive bibliographic certification.

7 Human-review recommendation

Recommended verdict: likely valid candidate counterexample. Review should focus on the two restrictions that extract a right inverse and on the scope distinction between Problem 1.5 and Conjecture 1.6.

References

- [1] M. Messerschmidt, *Strong Klee–Andô Theorems through an Open Mapping Theorem for cone-valued multi-functions*, arXiv:1606.00249v2, 2018.
- [2] M. Messerschmidt, *A family of quotient maps of ℓ^∞ that do not admit uniformly continuous right inverses*, arXiv:1909.10417v1, 2019.